SIMATS ENGINEERING
ASSESSMENT TOOL 3: Debugging

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED
CO2: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)

Assessment Tool Weightage: 25%

Question Count: 10 **Total Marks: 20**

Code of Conduct
I _____S. V. K. Sharon Surya(192324246) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________________

Debugging

Criteria	Problem Identification (4 Marks)	Debugging (4 Marks)	Optimization (4 Marks)	Analysis (4 Marks)	Code Quality (4 Marks)
Marks					

Total Marks Awarded:

Question 1: SQLite Database Connection

CODE:

```
import sqlite3


conn = sqlite3.connect("student.db")
```

```
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS Student(
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
    Department TEXT
)
""")

conn.commit()
conn.close()

print("Table Created Successfully")
```

Output

```
Table Created Successfully
```

Question 2: SQLite Record Insertion

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS Student")

cursor.execute("""
CREATE TABLE Student(
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
```

```
    Department TEXT
)
""")

cursor.execute("""
INSERT INTO Student(Student_ID, Name, Department)
VALUES(101,'Alice','CSE')
""")

conn.commit()

print("Record Inserted Successfully")

conn.close()
```

Output

```
Record Inserted Successfully
```

Question 3: SQLite Data Retrieval

```
import sqlite3

conn = sqlite3.connect("company.db")
cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS Employee")

cursor.execute("""
CREATE TABLE Employee(
    Employee_ID INTEGER PRIMARY KEY,
    Employee_Name TEXT,
    Department TEXT
```

```
)
""")

cursor.executemany("""
INSERT INTO Employee VALUES(?,?,?)
""",[
(1,"Rahul","CSE"),
(2,"Anitha","ECE"),
(3,"John","IT")
])

conn.commit()

cursor.execute("SELECT * FROM Employee")

rows = cursor.fetchall()

for row in rows:
    print(row)

conn.close()
```

Output

```
(1, 'Rahul', 'CSE')
(2, 'Anitha', 'ECE')
(3, 'John', 'IT')
```

Question 4: SQLite Record Update

```
import sqlite3

conn = sqlite3.connect("college.db")
```

```python
cursor = conn.cursor()

cursor.execute("""
UPDATE Student
SET Department=?
WHERE Student_ID=?
""",("ECE",101))

conn.commit()

print("Record Updated Successfully")

conn.close()
```

Output

```
Record Updated Successfully
```

Question 5: SQLite Record Deletion

```python
import sqlite3

conn = sqlite3.connect("bank.db")
cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS Customer")

cursor.execute("""
CREATE TABLE Customer(
    Customer_ID INTEGER PRIMARY KEY,
    Customer_Name TEXT
)
""")
```

```
cursor.executemany("""
INSERT INTO Customer VALUES(?,?)
""",[
(101,"Rahul"),
(105,"Arun"),
(110,"David")
])

conn.commit()

cursor.execute("""
DELETE FROM Customer
WHERE Customer_ID=?
""",(105,))

conn.commit()

print("Record Deleted Successfully")

conn.close()
```

Output

Record Deleted Successfully

Question 6: SQLite Record Retrieval

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()
```

```
cursor.execute("""
SELECT *
FROM Student
WHERE Student_ID=?
""",(101,))

row = cursor.fetchone()

if row:
    print(row)
else:
    print("Student Not Found")

conn.close()
```

Output

(101, 'Alice', 'ECE')

Question 7: SQLite Table Creation

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS Student")

cursor.execute("""
CREATE TABLE Student(
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
    Department TEXT
```

```
)
""")

conn.commit()

print("Student Table Created Successfully")

conn.close()
```

Output

Student Table Created Successfully

Question 8: SQLite Record Insertion

Aim

To insert a new employee record into an SQLite database.

Theory

The INSERT statement is used to add new records to a table.

Code

```
import sqlite3

conn = sqlite3.connect("company.db")
cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS Employee")

cursor.execute("""
```

```
CREATE TABLE Employee(
    Employee_ID INTEGER PRIMARY KEY,
    Name TEXT,
    Salary REAL
)
""")

sql = """
INSERT INTO Employee(Employee_ID, Name, Salary)
VALUES (?, ?, ?)
"""

cursor.execute(sql, (101, "John", 55000))

conn.commit()

print("Record Inserted Successfully")

cursor.close()
conn.close()
```

Output

Record Inserted Successfully

Conclusion

The program successfully inserts a new employee record into the SQLite database.

Question 9: SQLite Record Update

Aim

To update the salary of an employee in an SQLite database.

Theory

The UPDATE statement modifies existing records in a table.

Code

```
import sqlite3

conn = sqlite3.connect("company.db")
cursor = conn.cursor()

cursor.execute("""
UPDATE Employee
SET Salary=?
WHERE Employee_ID=?
""", (65000, 101))

conn.commit()

print("Record Updated Successfully")

cursor.close()
conn.close()
```

Output

Record Updated Successfully

Conclusion

The program successfully updates the employee salary.

Question 10: SQLite Record Deletion

Aim

To delete all customer records belonging to Chennai.

Theory

The DELETE statement removes records from a table based on a condition.

Code

```
import sqlite3

conn = sqlite3.connect("bank.db")
cursor = conn.cursor()

cursor.execute("DROP TABLE IF EXISTS Customer")

cursor.execute("""
CREATE TABLE Customer(
    Customer_ID INTEGER PRIMARY KEY,
    Customer_Name TEXT,
    City TEXT
)
""")

cursor.executemany("""
INSERT INTO Customer VALUES(?,?,?)
""",[
(101,"Rahul","Chennai"),
(102,"Alice","Delhi"),
(103,"John","Chennai")
])
```

```
conn.commit()

city = "Chennai"

query = "DELETE FROM Customer WHERE City=?"

cursor.execute(query, (city,))

conn.commit()

print("Records Deleted Successfully")

cursor.close()
conn.close()
```

Output

Records Deleted Successfully